

Answer the questions. Students' answers should be based on the texts and videos from the main course books (Essential English for IT students. Part 2)

1. What types of wired internet connections do you know? Talk about each of them in detail. (Unit 1_the Internet, text A)
2. What is the difference between DSL and dial-up connection?
3. Talk about the three wireless internet access technologies mentioned in the text. (Unit 1_the Internet, text A).
4. What are the three types of DSL connections that you know? Talk about each of them. (Unit 1, text A, video task 5)
5. What is the difference between packet switching and circuit switching technologies?
6. What are the two subtypes of packet switching technologies that you know? Talk about each of them. (Unit 1_the Internet, text B)
7. Talk about the TCP/IP model (the layers, protocols).
8. What is the difference between TCP and UDP?
9. What is the difference between the Internet and the WWW?
10. Talk about the three types of search engines and their distinctive features. (Unit 2, Search Engines, text A)
11. What are the three steps that every crawler-based search engine takes before displaying search results? Talk about each of these stages. (Unit 2, Search Engines, text A)
12. What makes meta-search engines different from other types of search engines?
13. What is the difference between crawler-based search engines and human-powered directories?
14. What factors determine higher ranking in search results? What aspects can ranking algorithm take into consideration? (Unit 2_Search Engines, video task 5)
15. What is the difference between a search engine and a web browser?
16. Talk about any four search engines that can be used as an alternative to Google. (Unit 2_Search Engines, text B)
17. How can programming languages be classified? Provide examples and talk about the distinctive features of each type you have named.
18. What is OOP? What are the three key features of OOP? Talk about each of them in detail. (Unit 3_Programming languages, text D)
19. What is the difference between a compiler, an interpreter, and an assembler?
20. What are the three main tools used in web design? Talk about each of them. (Unit 4_Web design, text A)
21. What is the difference between frontend and backend web development?
22. What are the advantages and disadvantages of using website builders?
23. What is SEO? What is its main purpose?

24. What is the difference between on-page SEO and off-page SEO? (Unit 2_Search Engines, video task 8)
25. What is meant by “organic results”? (Unit 4_Web Design, task 8)
26. What is malware? What types of malware do you know? Talk about each one in detail. (Unit 5_Malware, text A)
27. What are the four routines of a virus? Talk about the function of each of them. (Unit 5_Malware, text D)
28. What types of cybercrime do you know? Talk about each of them. (Unit 5_Malware, text A)
29. What is the difference between a virus and a worm?
30. What is the difference between a virus and a Trojan?
31. What is the difference between DoS and DDoS attack? How is DDoS attack performed?
32. What is firewall? What can firewall filters be based on? (Unit 6_Data Security, video task 7)
33. What is the difference between signature-based scanning and heuristic scanning?
34. What is a signature? (Unit 6_Data Security, text B)
35. What are the two main types of encryption? What is the difference between the two? (Unit 6_Data Security, text C)
36. What two methods of breaking a cipher are there? (Unit 6_Data Security, text C)
37. What is a CAPTCHA? Where is this tool implemented?
38. What is the Turing test? Talk about its procedure.
39. What is the difference between classification and clustering? (Unit 7_the Future of IT, text A)
40. What is backpropagation?
41. What is an artificial neural network? What tasks can neural networks accomplish?
42. In what way is an artificial neural network similar to biological brain? Describe the structure of artificial neural networks.
43. Why are neural networks compared to ‘black boxes’?
44. What is IoT? What are the major problems in using the IoT technology? (Unit 7_the Future of IT, text B)
45. What is HLAI? What are the risks of using HLAI?